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The impact of renewable energy, internet use and foreign direct investment on carbon dioxide emissions: A method of moments quantile analysis

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A B S T R A C T

The current era of industrial development and innovation is revolutionized using the internet, robotics, and artificial intelligence. Nevertheless, the appraisal of global economic progress shows increasing trends, there are environmental degradation issues associated with this improvement. The role of renewable energy, urbanization and foreign direct investment received a lot of attention in the literature on environmental issues, however, the simultaneity with information communication technology is missing. Therefore, the present study used data from 10 emerging countries during 1996–2015 and applied the novel Method of Moments Quantile Regression to analyze the nexus among the variables. Further, we applied the second-generation unit root test, and Driscoll Kraay standard errors to reach robust results. The findings revealed an inverted U-shape relationship between economic growth and environmental degradation; thus, the validity of the Environmental Kuznets Curve is revealed. Moreover, foreign direct investment is significant and positive at 0.05th-0.50th quantiles, however, it becomes insignificant at higher quantile levels. Urbanization enhances while renewable energy mitigates carbon dioxide emissions at all quantile levels. Information communication technology proxied by internet usage reduces environmental degradation significantly at 0.25th-0.95th quantile levels. Results of the study suggest insights for the policymakers to mitigate carbon dioxide emissions through encouraging renewable energy and internet use.

1. Introduction

The theoretical environmental concerns have become a real-world issue for recent and future generations. The massive progress, increase in production and consumption of industrial output caused such environmental degradation. Among various greenhouse gases, carbon dioxide (CO_2) emissions is the major contributor and is now the target of different global environmental agendas. Earlier literature has discussed various factors that impact CO_2 emissions, including renewable energy use, economic growth, financial development, urbanization [1–5]. Perhaps renewable is the most plausible solution to curb CO_2 emissions. Nevertheless, the impact of energy consumption on the environment is damaging and reducing energy consumption could be a disaster for economic growth. However, renewable energy is an environmental-friendly energy source. Renewable energy is promising in a sense that it is supportive to the environment, low waste production is involved in its generation, and it ensures reliable energy supply. Previous

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literature argued that achieving environmental sustainability is impossible without promoting renewable energy [6–9]. Earlier literature reported Renewable energy reduces CO_2 in the EKC framework [10,11].

On the other hand, one of the primary ingredients of economic growth is the Information and communication technologies (ICT) that accelerate economic growth [12,13]. The Internet, mobile, telephone, computer systems, and associated applications, collectively known as ICTs, have become the primary drivers of societal transformation, growth, and invention. On the other hand, the broad usage of ICT contributes to increased production efficiency. In addition, ICT has contributed to significant systemic transformations in the international economy over the last few decades by building smart cities, health systems and transportation systems [14] and through encouraging global supply chains and boosting economic development [15].

While ICT is regarded as a stimulus to economic growth, and subsequently, its environmental impacts are keenly investigated. ICT has been recognized as a vital factor to achieve a low carbon economy [15]. Earlier research argued that ICT might have a good or detrimental impact on environmental stability. ICT, for instance, improves environmental protection by lowering CO_2 emissions [16]. Even though [17] claimed that Internet commerce reduces CO_2 emissions in general. The impact of the Internet on CO_2 emissions, on the other hand, varies significantly between industrialized and developing nations.

Similarly, ICTs aid in the mitigation of atmospheric greenhouse gases (GHG) in two directions: by boosting the productivity of existing procedures and by causing alternative consequences that occur in more energy-efficient output and utilization processes [18]. In general, ICTs are one of the causes that contributes to accelerated CO_2 emissions through the manufacture of ICT gear and devices, energy use, and electronic material recycling. Moreover, ICT use enhances energy demand, as documented by Refs. [19,20].

Furthermore, urbanization may have a positive or negative role in CO_2 emission [21], as discussed by urbanization theories, including modernization theory and environmental transition theory. Earlier literature shows that with ICT, urbanization becomes more efficient and fast [22]. With the rise of urbanization, demand for energy increases due to electronic devices and automation machines usage [23], therefore, energy, urbanization and internet usage are linked [20,24]. However, their environmental consequences require a careful analysis [25].

In addition, the influx of FDI may result in increased CO_2 emissions in the host nation, particularly in states that desperately seek economic prosperity and a loosening of environmental restrictions to lure foreign capital. On the other hand, FDI inflows may result in beneficial knowledge transfer and headwinds, efficient energy use, and lower CO_2 emissions in host nations. Thus, understanding the intricate connections between FDI inflows and CO_2 emissions is critical for providing better policy guidelines to these issues.

Previous literature shows that the link between renewable energy, ICTs and CO_2 emissions is conflated and needs more consideration. Hence, this study aims to further investigate the impact of ICTs on CO_2 emissions in emerging economies by integrating FDI, urbanization, GDP, GDP^2, and renewable energy into the EKC framework. The global percentage of CO_2 emissions emitted by emerging economies increased from 48% in 1980 to 61% in 2011 [26]. In addition, previous literature utilized the STIRPAT model to analyze ICT, urbanization and CO_2 nexus [25]. However, our study used the EKC framework. Thus our study will contribute to the debate initiated by Ref. [27] on EKC vs STIRPAT efficiency.

This research adds to the thin body of literature on the nexus above using the MMQR [28]. Combining this technique with fixed effects makes it simpler to gain a practical grasp of the association's heterogeneity. This technique allows for diverse environmental-emissions linkages at multiple conditional quantiles distributions, not captured by traditional mean regressions—panel estimate methods, including the novel "Method of Moments Quantile Regression" (MMQR). In addition, the fully modified ordinary least square (FMOLS) is utilized with the dynamic ordinary least square (DOLS), which accounts for the endogeneity concerns in cross-sectional data, and the Dumitrescu Hurlin panel causality are other unique features of the analysis. The estimation findings provide policy implications based on the nexus between FDI, ICT, renewable energy, and environmental policy for emerging countries. In addition, Figs. 1–6 shows the recent trends of FDI, ICT, economic growth, urbanization, environmental degradation, and renewable energy in 10 emerging countries. China and India show a solid upward trend in environmental degradation and economic growth. In the sample period, Turkey, Hungary, and India attracted considerable FDI. The process of urbanization in China remained remarkable during this period. The share of renewable energy in total energy consumption increased in Hungary, Romania, and Bulgaria. Internet usage shows an increasing trend in all countries.

The research contributes to the environmental literature in a variety of ways. To our best knowledge, this is the first research of its kind that rigorously assesses the role of renewable energy, FDI, ICTs, and urbanization in CO_2 emissions for emerging countries. For emerging economies, the nexus between ICT, urbanization, and renewable energy with the EKC hypothesis has not received due attention, therefore, extending the previous literature with more advanced econometric techniques is required for a detailed analysis.

Following the introduction, the body of the article is organized as follows: section 2 “literature review,” section 3 “data model and econometric approaches” section 4 “result and discussions” last section is conclusions and policy implications.

2. Literature review

Earlier literature has linked CO2 emissions to a variety of factors including GDP, financial development, nonrenewable and renewable energy use [1–5]. Before analyzing the dynamics of carbon dioxide emissions, renewable energy consumption with an emphasis on internet usage FDI, economic growth and urbanization, it is vital to discuss the scientific contribution related to this theme in the previous literature. The extant literature related to our study can be classified into three strands.

The first strand investigated the relationship between economic development and environmental pollution. This strand mainly follows [29] to examine the environmental impact of economic development. This environment and development nexus was named the Environmental Kuznets Curve (EKC) after the famous study by Simon Kuznets [30]. Later a plethora of research followed the EKC hypothesis to study the environment and income growth relationship [31–35]. Interestingly, this strand has two contrasting conclusions; on one hand, studies are promoting the idea of the EKC hypothesis [36,37]. However, on the other hand, there are findings by others that support the idea of invalidity of the EKC hypothesis [27,38–47].

The second group of the literature analyzed the linkage between renewable energy and economic growth. Some studies found bidirectional causality among renewable energy and economic development [48,49]. In a recent study, renewable energy consumption and economic development are reported to have a bidirectional causality for a sample of 58 countries [50]. In contrast, some other studies reported no causal association concerning economic growth and renewable energy, for instance Ref. [51] found no causality in a sample of European countries. One can find

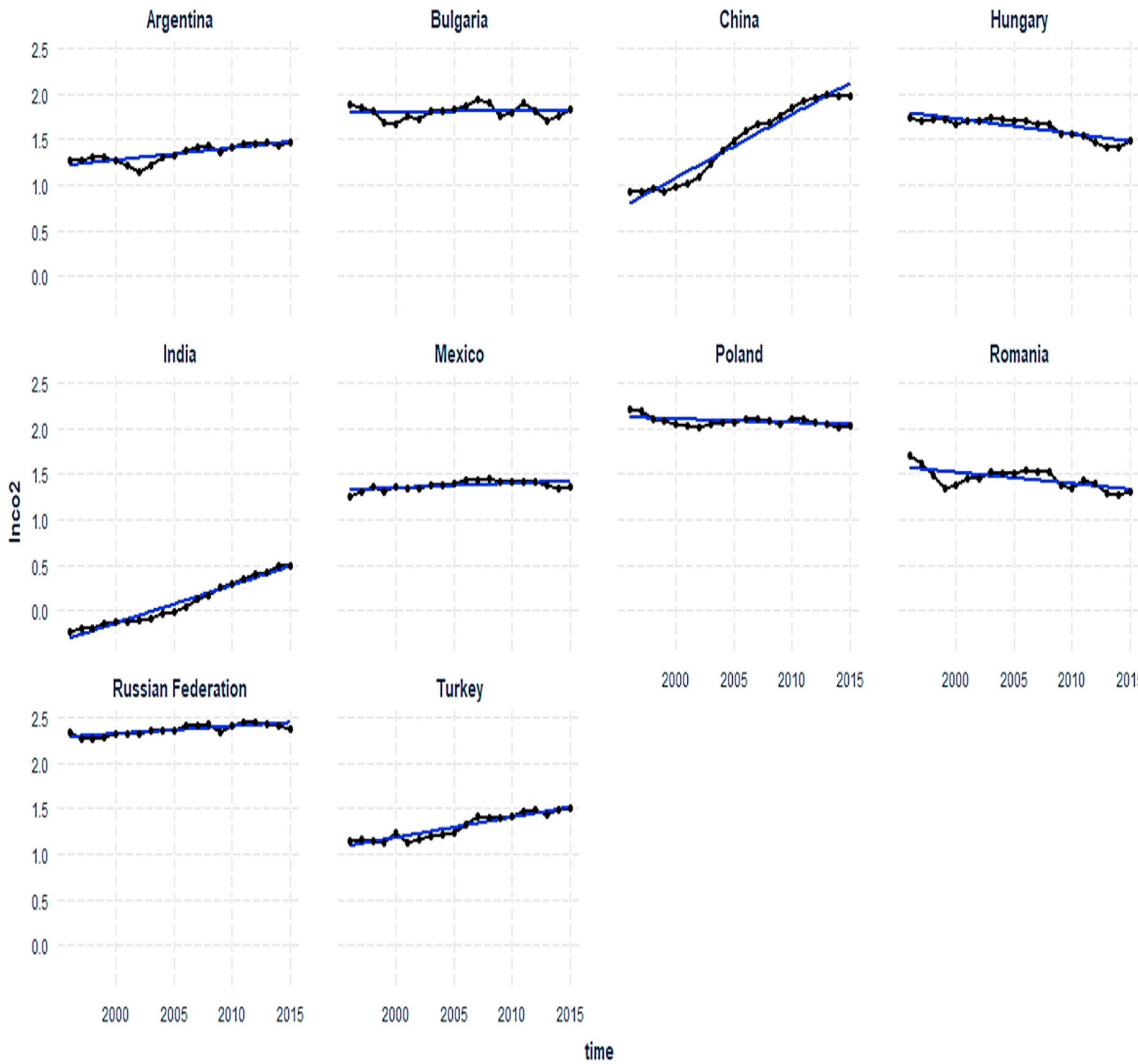

Fig. 1. Trends in carbon dioxide emission.

in previous literature, that unidirectional causality is reported also [52].

Besides, economic growth and renewable energy, the EKC literature also employed other aspects of economies. For instance, FDI, urbanization and internet usage were found by previous literature as explanatory variables of environmental pollution. Literature on urbanization and environmental degradation can be broadly viewed into three distinct groups. The first reported a positive effect of urbanization on environmental degradation [53]. As more population shifts from rural to urban area, demand for energy increases due to the rise in vehicles and residential buildings. This results in the deterioration of the environment. Environmental transition theory justifies such linkage between environment and urbanization. In a recent study, urbanization is found to degrade the environment in a sample of 99 countries [54]. However, the notion of agglomeration disagrees with the increasing effect of urbanization process on the environment and argues for a mitigating effect. As the income level rises and urbanization increases in an economy, fewer resources are required for transportation and accommodation. Therefore, due to urban agglomeration, the consumption of energy reduces [55,95,96]. In addition, ecological modernization theory also supports this idea of improvement in the environment at later stages of development due to efficient technologies [53].

The current study also included FDI in the model as an independent variable. FDI has been a central focus of studies targeting the environment and economic development nexus. Foreign direct investment in many of the developing countries is attracted by weak environmental regulations. Particularly, countries that are rich in natural resources with weak environmental regulatory laws are attracting FDI [56]. Literature on the nexus of FDI, Economic growth and environment can be classified into two major groups. First; studies that found a positive effect of FDI on the environment called the “pollution halo effect” [57]. And the second group that found FDI mitigating environment degradation, called the “pollution heaven effect” [58]. In a recent study [59], analyzed the role of FDI and ICT on CO_2 for Pakistan during 1990Q4-2018Q1 using the NARDL model. The findings of this study suggested that local production of ICT equipment improves the environment as compared to importing them. Furthermore, their study endorsed the validity of the EKC for Pakistan. The study found a positive effect of FDI on environmental degradation in the long run.

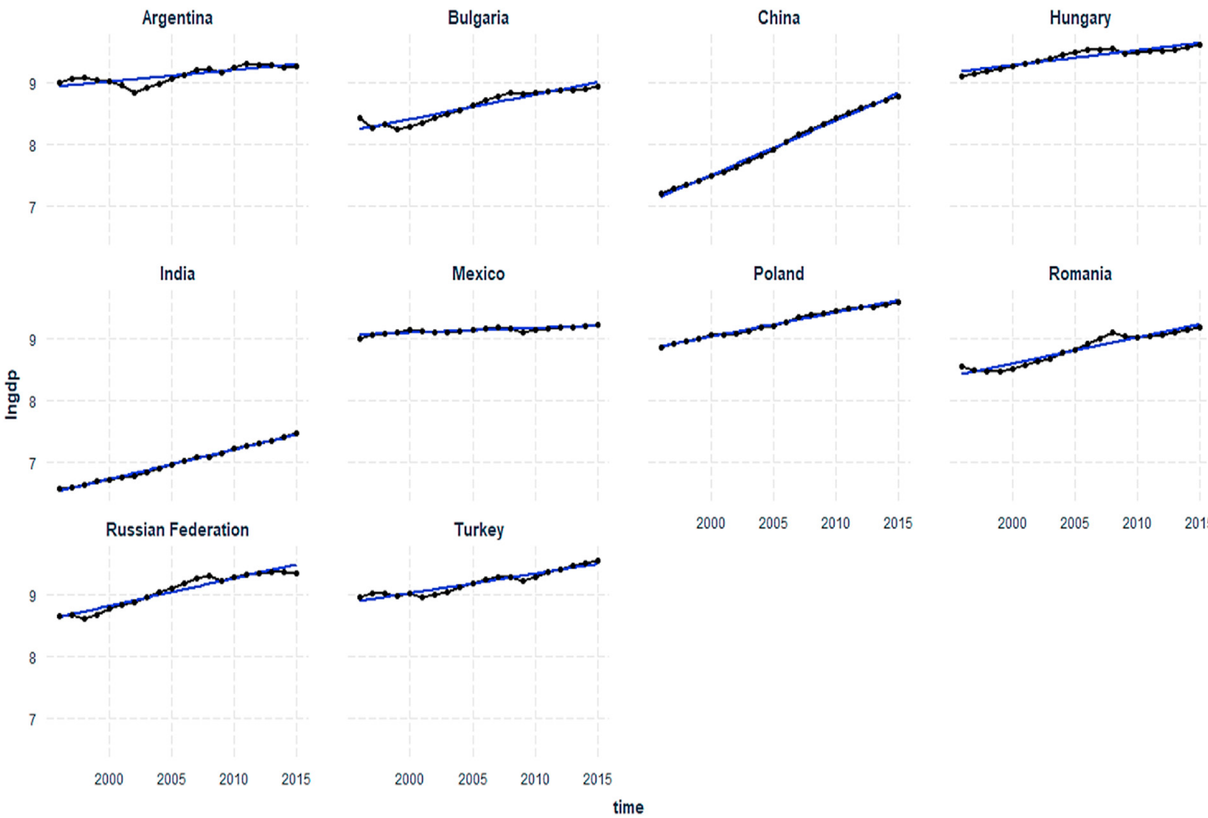

Fig. 2. Trends of gross domestic product.

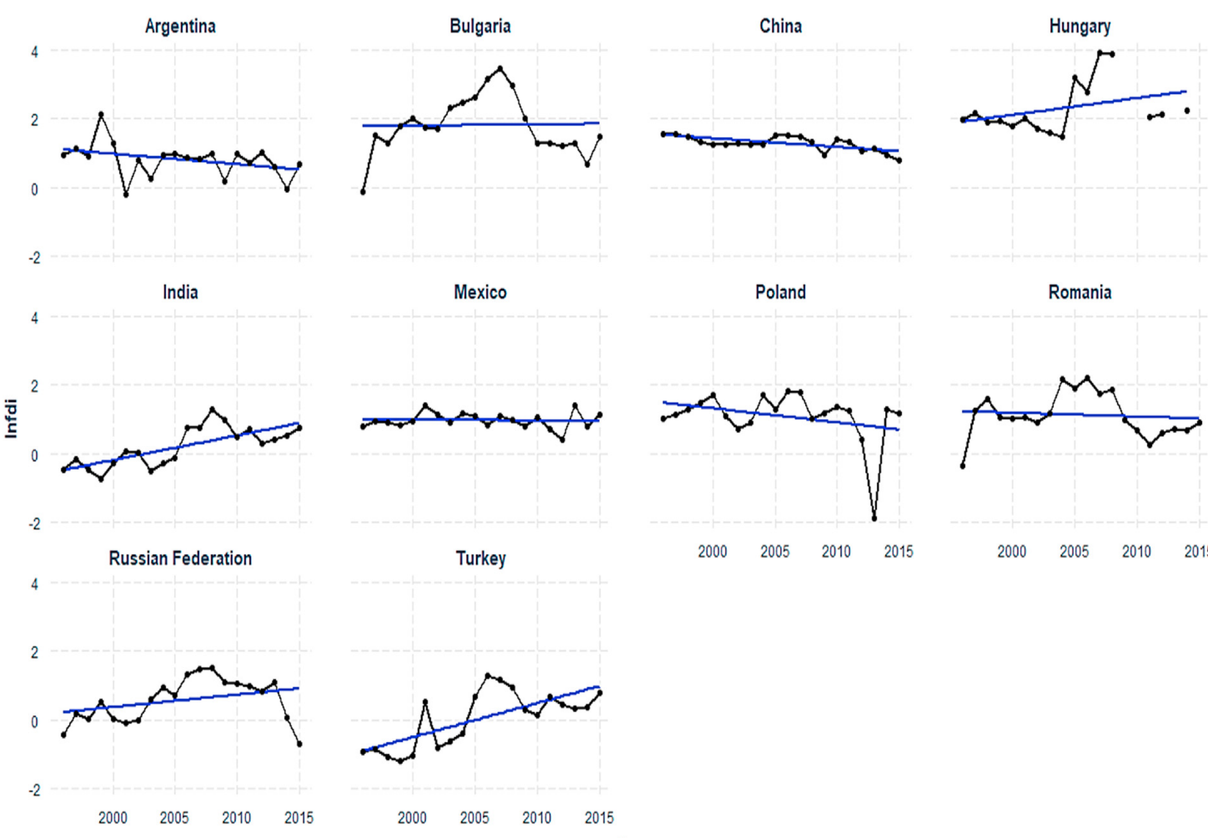

Fig. 3. Trends in foreign direct investment.

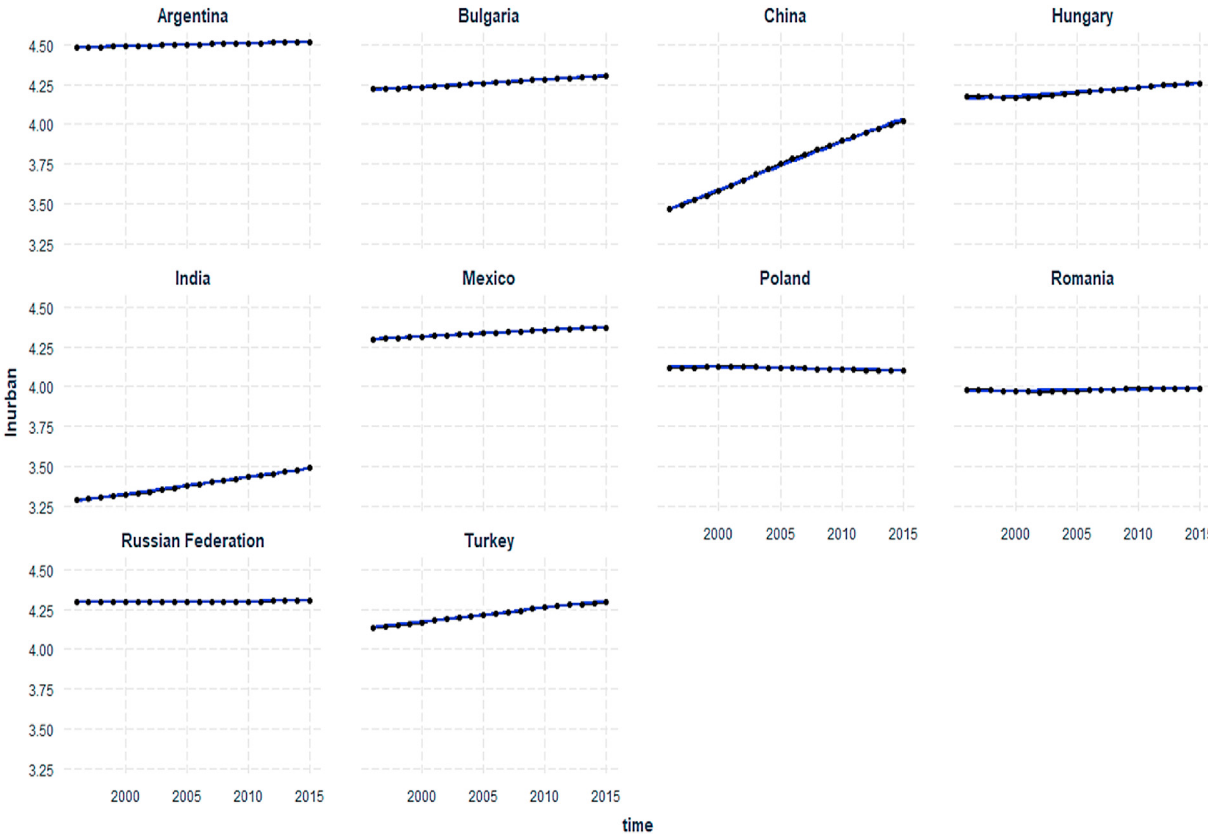

Fig. 4. Trends in urbanization.

Among many technologies, the internet is perhaps the most effective technology that reshapes societies, particularly in urban areas. One of the earlies studies on the role of information technology was [60]. This study offered a new perspective in environmental analysis and introduced information technology as a new determinant. Later [61], offered a detailed framework for further investigations about the impact of ICT in sustainability. In the same vein [62], analyzed the role of e-commerce on the environment and found a positive impact of former on later. Similarly [63], found that reading online newspapers reduces environmental degradation. Later studies also found environmental benefits of ICT, for instance Refs. [64–66] through energy efficiency [67], by reducing energy demand [68,69] energy saving by using GPS in the transport sector [70], through enhancing online retailing. However, the extant literature is conflated about the role of internet usage and other forms of ICT in environmental degradation. On the one hand, there are empirical findings that supports the idea of a harmful effect of ICT on environmental degradation [63,64,68]. While, other studies argue that an increase in internet users, increase demand for more electricity and electricity production puts pressure on the environment [71,72].

Thus from the literature review, three gaps are visible; firstly there is no study on the role of ICT, FDI and urbanization in the EKC framework, Secondly, in the case of emerging economies, there is no study on the emission-growth nexus, particularly concerning ICT and urbanization. Thirdly, with a focus on cross-sectional dependence, the application of advanced models that analyze the heterogeneous effects of independent variables on various levels of CO_2 emissions is a key gap in the extant literature on the EKC hypothesis.

From the detailed review of literature presented above, it is evident that there is vitally important to guide policymakers about the relationship between renewable energy, FDI, internet usage and urbanization with the EKC in emerging economies.

3. Data and methodology

For empirical investigation, the current study considers 20 years of annual time series data spanning from 1996 to 2015 for constructing a balanced panel of 10 emerging economies (Argentina, Bulgaria, China, Hungary, India, Mexico, Poland, Romania, Russian Federation, and Turkey). We have included these 10 emerging economies in our present study as they have surpassed the other emerging economies CO2 emissions within the sample period. Moreover, the choice of sample selection is purely based on data availability and for the purpose of a balanced panel data. The multivariate framework of the present study utilizes CO_2 emissions to capture environmental degradation measured in metric tons per capita, real GDP as a proxy of economic well-being measuring in per capita, consumption of renewable energy measured as percentage of total consumption, foreign direct investment captured through net inflows in terms of percentage of GDP, urbanization (percentage of the total population) and individuals using the internet as a proxy of internet penetration measuring in percentage of the total population. The data source for all the series involved in the study are retrieved from the World Bank. Each of the variables has gone through normal logarithmic change to actuate stationarity properties and stay away from the issues of distributional properties. The comprehensive description of variables is enlisted in Table 1.

Table 2 uncovers that all the variables follow left-skewed distribution, except renewable energy consumption and FDI following right-skewed distribution. All the variables except renewable energy consumption are demonstrated not to follow a normal distribution as evidenced from the Jarque-Bera test. These results

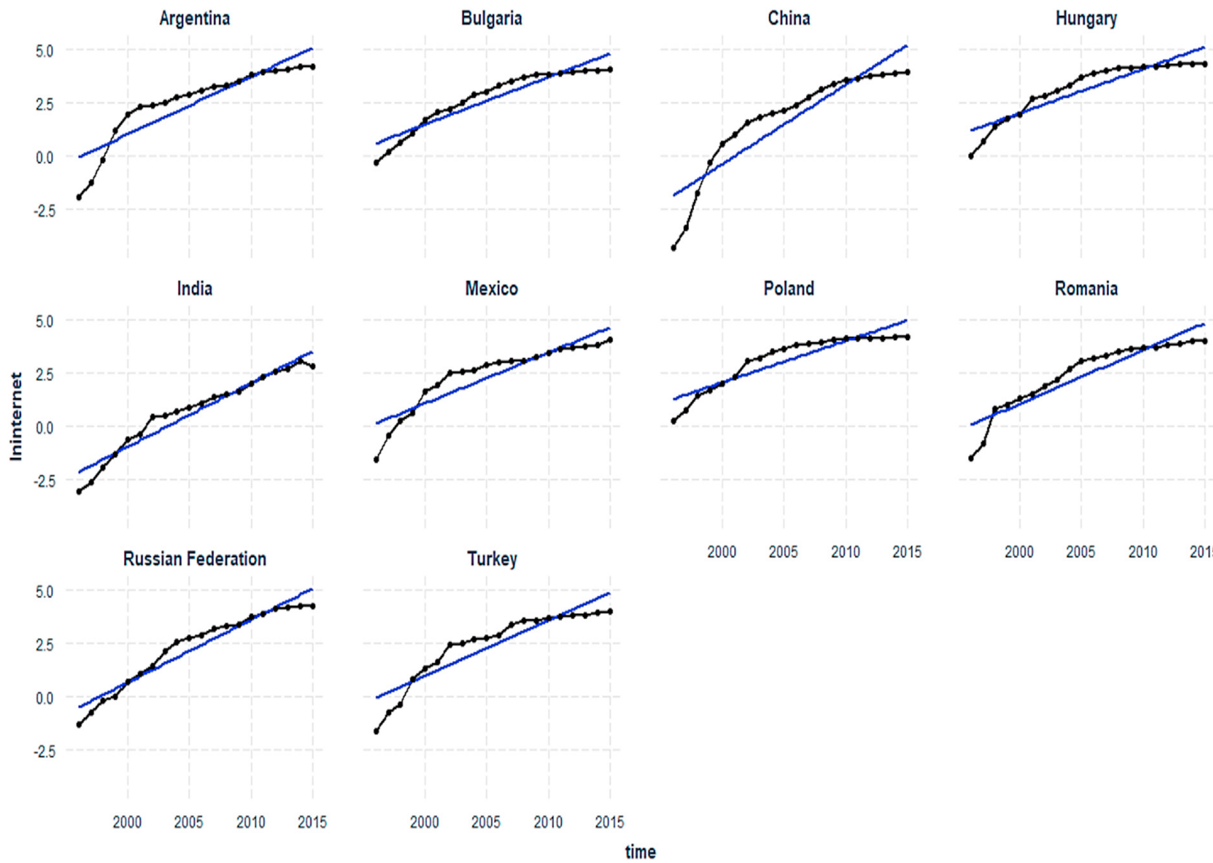

Fig. 5. Trend in Internet users.

support the idea of using nonlinear models that can capture the heterogenous relationship among the factors of dependent variable.

3.1. Panel estimation techniques

Concocted by Machado and Silva [28], the present study employed MMQR model to investigate the heterogeneous and distributional impacts across quantiles among CO2, real GDP growth, renewable energy, FDI, urbanization and internet usage. While conventional panel quantile regression techniques [73,74] can provide reliable estimates in the presence of outliers and appropriate in a situation when conditional means of two variables have a weak association, however, they are unable to detect the possible presence of unnoticed heterogeneity across the panel cross sections. Authorizing the individual fixed effect, MMQR method unavoidably examines the conditional heterogeneous covariance impact of CO_2 emission's determinants to effect the whole distribution. In addition, rather shifting means like Koenker [73], Canay [74], it captures the covariance effect in overall distribution. The advantage of utilizing this method is that it addresses the conceivable presence of endogenous properties in the explanatory variables. This technique is likewise appropriate in such situations where individual effects submerge the panel data model. Moreover, the MMQR model also produces reliable estimates in the case of the non-linear model. As compared to the nonlinear models such as “Nonlinear Autoregressive Distributed Lag” NARDL, the advantage of using MMQR is that it defines the threshold through data driven process not exogenously [75]. Moreover, MMQR allows for asymmetries based on location. MMQR can also provide huge bits of knowledge in the event of creating non-crossing estimates in quantile regression. Although fixed effects cannot account for heterogeneity, yet the application of MMQR covers this issue due to its ability to render heterogeneous estimates across the entire distribution. Moreover, the heterogeneous values of coefficients explicitly claim that MMQR deals with the issue of heterogeneity. Conducting error analysis (using Mean Square Error) Machado and Silva [28] showed through simulation results that, MMQR method provides more robust estimates as compared to conventional models. Other methods of panel regression such as FMOLS and DOLS have advantage of dealing with serious correlation and endogeneity, however, they also lack from estimation based on condition in the data. Therefore, the method is preferred because it deals with heterogeneity and endogeneity by considering asymmetric and non-linear association among CO_2 emissions and its determinants. The conditional quantile estimates $Q_y(\tau|X)$ of the model of location-scale variant can be expressed by the following equation:

$$y_{it} = \alpha_i + X_{it}{'}\beta + (\delta_i + Z_{it}{'}\gamma)U_{it} \tag{1}$$

Where, the probability P $\{\delta_i + Z'_{it}\gamma > 0\} = 1$, $(\alpha, \beta', \delta, \gamma')'$ are the parameters to be estimated. The individual i fixed effects are designated as (α_i, δ_i), $i = 1, \ldots,$ n and k vector of known elements of X is signified by Z, which are differentiable conversions with component l mentioned below:

$$Z_l = Z_l(X)\ l = 1, \ldots, \ k \tag{2}$$

X_{it} is independently and identically distributed for any fixed i and also across time t. U_{it} is also independently and identically

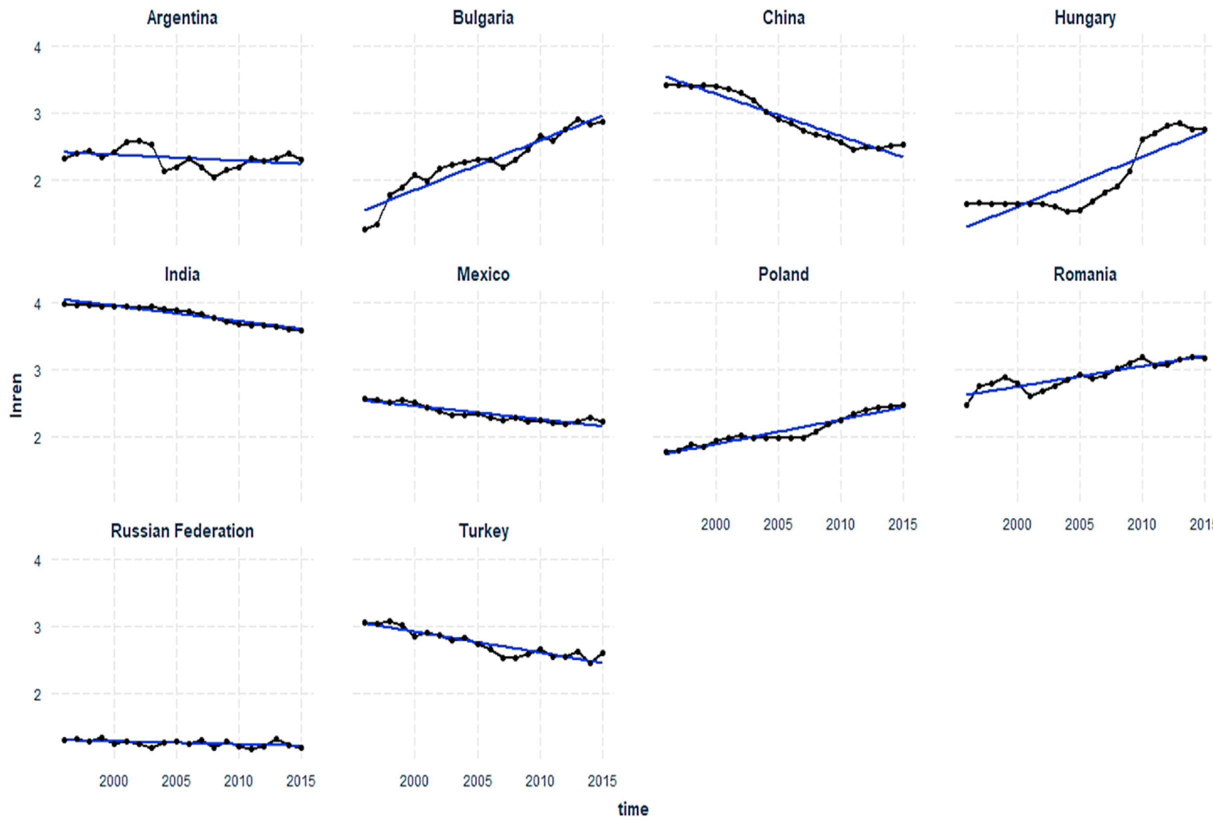

Fig. 6. Trends in renewable energy.

Table 1
Description of variables.

Variable	Description of variables	Measurement	Source
LnFDI	Foreign direct investment	Net inflows (% of GDP)	WDI
LnUrban	Urban population	Urbanization (% of total population	WDI
LnCO2e	Environmental pollution	Carbon emission (mto in per capita form)	WDI
LnGDP	Economic Development	GDP in US dollars (constant-2010)	WDI
LnREN	Renewable Energy	As percentage in total energy cons.	WDI
LnInternet	Internet penetration	Individuals using internet (% of population)	WDI

Note: WDI=The World Bank.

Table 2
Descriptive statistics.

Variables	lnCO2e	lnGDP	lnGDPsq	lnren	lnfdi	lninternet	lnurban
Mean	1.492	8.756	77.224	2.467	0.986	2.328	4.105
Std. Dev.	0.598	0.752	12.361	0.708	0.884	1.798	0.316
Min	−0.235	6.568	43.138	1.172	−1.88	−4.337	3.289
Max	2.454	9.606	92.271	3.985	3.921	4.326	4.516
Skew.	−0.993	−1.468	−1.327	0.17	0.072	−1.188	−1.189
Kurt.	4.195	4.181	3.799	2.677	4.368	3.955	3.584
JB	44.3	83.2	63.05	1.6	15.4	54.1	49.5
Probability	0.000	0.000	0.000	0.423	0.000	0.000	0.000
Observations	200	200	200	200	200	200	200

distributed across individuals *i* through time t and are orthogonal to X_{it} and are standardized to accomplish the moment conditions. Equation (1) derives the following:

$$Q_y(\tau|X) = (\alpha_i + \delta_i(\tau)) + X'_{it}\beta + Z'_{it}\gamma q(\tau) \quad (3)$$

Where, x'_{it} signifies a vector of independent variables that augmented GDP, GDP^2, renewable energy, FDI, urbanization and internet. $Q_y(\tau|X)$ postulates that the structural quantiles are distributed to the dependent variable y_{it}(CO_2 emissions) relying on the distribution (location) of exogenous variables X_{it}. Individual (i) quantile (τ) fixed effects is demonstrated by the scalar coefficient denoted as $\alpha_i(\tau) = \alpha_i + \delta_i q(\tau)$. Intercept shift does not represent the individual effect in contrast to the typical fixed least-squares effects. Such parameters are time-invariant with heterogeneous effects that are suitable to diverge along the conditional distributional quantiles of the endogenous variable. The τ-the sample quantile represented by q(τ) can be assessed by addressing the resulting optimization problem written in equation (4):

$$min_q \sum_i \sum_t \rho_\tau (R_{it} - (\delta_i + Z'_{it}\gamma)q) \tag{4}$$

Where $\rho_\tau(A) = (\tau - 1)AI\{A \leq 0\}TAI\{A > 0\}$ indicates the check function.

4. Empirical results

4.1. Testing unit root

The stationary properties of the variables are evaluated using second-generation unit root tests to avert the problem of spurious regression in subsequent estimations. The present study applied panel unit root test by Pesaran [76] and outcome is presented in Table 3. The Pesaran [76] panel unit root test (CIPS) is conducted in levels and first difference. Results show that only FDI and ICT are non-stationary at levels however, all the series are stationary at the first difference, thereby signifying that all the variables are either I(0) or I(1).

4.2. Panel cointegration test

To identify the long-term cointegrating relationship between the variables, subsequently, the bootstrapped panel cointegration test of Westerlund [77] is employed. The results are presented in Table 4. The bootstrapped panel cointegration test of Westerlund [77] shows evidence of long term cointegrating relationship. The null hypothesis of no cointegrating relationship is rejected at a 1% level of significance for all the tests. The robustness of the cointegration relationship is confirmed by the Kao [78] test for panel cointegration. It can be observed that at 1% level of significance the alternative hypothesis is accepted. The findings reveal that the variables have a long-term association, verifying the validity and consistency of empirical findings.

4.3. Panel regression estimates

The results of panel regression models are presented in Table 5 using FMOLS and DOLS. Furthermore, to reach robust results, subsequently, we estimated FE-OLS model with robust Driscoll-Kraay standard errors are presented in Table 6. It can be observed that coefficients of GDP and GDPsq are positive, and negative respectively, thereby suggesting the existence of “Inverted U-shape EKC” for the panel of 10 countries for the study period of 1996–2015. Further, a 1% increase in FDI and Urbanization contributes positively to the carbon emission by 0.008% and 1.244%, however, consumption of renewable energy reduces carbon emission by 0.29%. However, ICT shows evidence of a negative relationship with CO2e, whereby 1% increase in ICT reduces carbon emission by 0.037%. All the coefficients are significant at 1% level of significance. The robustness of the results is reviewed by excluding the ICT variable and presented in Table 6. It can be observed that coefficient signs and size are almost similar to the results presented in Table 5. Moreover, the results are also statistically significant at 1% level.

Table 3
Unit root test (Pesaran CADF).

Variables	Level	First difference
Lnco2	−0.882 (0.189)	−4.992*** (0.000)
Lnren	−0.473 (0.318)	−5.530*** (0.000)
Lngdp	−0.069 (0.473)	−3.593*** (0.000)
Lngdpsq	−0.050 (0.480)	−3.420*** (0.000)
Lnfdi	−0.144* (0.073)	−4.560*** (0.000)
Lnurban	7.165 (0.8600)	−2.506*** (0.000)
Lninternet	−1.537*** (0.006)	−3.450*** (0.000)

Table 4
Cointegration test Westerlund (2007).

Statistic	Value	Z-value	P-value
Gt	−3.222	−1.261	0.000
Ga	−4.239	2.508	0.994
Pt	−3.24	3.277	0.001
Pa	−3.493	1.134	0.872

Note: (H0): No cointegration; (Ha): at least one of the cross-sectional units has cointegration (Gt and Ga) or Cointegration for the panel as a whole (Pt and Pa).

4.4. Method of Moments Quantile Regression

The results for Method of Moments Quantile Regression (MMQREG) are presented in Table 7. The coefficient of renewable energy consumption (REC) is negative and statistically significant at a 1% level of significance in all quantiles. The coefficient value varies from −0.264 in 5th quantile to −0.2529 in 95th quantile, thereby suggesting that a 1% increase in renewable energy consumption aids in reducing the CO_2 emission (CO2e) by approximately 26% in emerging economies. It can be observed that the effect of economic growth on CO2e is positive and highly significant in all the quantiles. Moreover, the GDP square term (GDP^2) is negative and highly significant in all quantiles. Juxtaposing the results, the coefficients of GDP and its square are positive, and negative respectively, thereby suggesting the existence of “Inverted U-shape EKC” for the panel of 10 countries for the study period of 1996–2015. The coefficient of FDI is positive and statistically significant at a 10% level for the lower quantiles (5th, 25th' and 50th), thereby highlighting that a 1% increase in FDI increases CO2e by 0.0125%–0.02% for emerging economies. The coefficient of urbanization is significant and positively correlated with environmental degradation in all quantiles, thereby highlighting that a 1% increase in urbanization can deteriorate the environment by 0.81–0.83% across all the quantiles of carbon emission. Further, the coefficients are following a rising trajectory which indicates that higher CO2e quantiles are more affected due to urbanization-driven environmental degradation. However, the coefficients of ICT are negative and significant for all quantiles except for the 5th quantile and reveal that a 1% increase in ICT reduces carbon emission by 0.0125%–0.0254%. The coefficient size gradually becomes more negative from lower to higher quantiles. Fig. 7 also shows the graphical representation of coefficients for various factors at different quantile levels.

4.5. Discussion of the outcome

The findings of the method of moment and fixed effect regression, both indicate that internet usage has a statistically significant negative impact on CO2e, thereby indicating that internet usage contributes in the improvement of air quality for emerging economies. This findings is aligned with [79] for emerging countries [80], for sub-Saharan Africa countries [81], for BRICS economies. As emerging economies like India, China, Argentina etc. are witnessing a boom in internet penetration, the higher energy savings due to the substitution effect of ICT is surpassing the rebound effect and compensation effect of the additional energy demand of ICT adoption. Moreover, rising internet penetration encourages knowledge sharing and collaborative research with higher

Table 5
Panel estimations through FMOL and DOLS.

Variables	Dynamic OLS			Fully Modified OLS		
	Coef	S.E	T-value	Coef	S.E	T-value
Lnren	−0.291***	0.039117	−7.439700	−0.2564***	0.009042	−8.9032
Lngdp	2.129***	0.679635	3.132926	1.8059***	0.338631	5.3331
Lngdpsq	−0.106***	0.038101	−2.784618	−0.0857***	0.018860	−4.5446
Lnfdi	0.008*	0.011230	0.752823	0.0150*	0.006830	1.6620
Lnurban	1.244***	0.453905	2.742570	0.8110***	0.028803	3.8989
Lninternet	−0.037***	0.008855	−3.927561	−0.0170***	0.008011	−2.4992
R square				0.92		
Adj- R square				0.91		

Note: the sign (*) denotes the significance level, * at 10%, ** at5%, and *** at 1% level of significance.

Table 6
Driscol Karaay Regression results.

Variables	(1)	(2)
	$lnCO_2$	$lnCO_2$
lnren	−0.258***	−0.263***
	(0.0138)	(0.0157)
lngdp	1.785***	1.770***
	(0.195)	(0.153)
lngdpsq	−0.0847***	−0.0887***
	(0.0107)	(0.00829)
lnurban	0.818***	0.783***
	(0.159)	(0.121)
lnfdi	0.0123***	0.00715*
	(0.00415)	(0.00404)
lninternet	−0.0167***	
	(0.00450)	
Constant	−10.28***	−9.721***
	(0.326)	(0.446)
Observations	196	196
R-sq	91.01	90.32

Note: the Driscoll Kraay standard errors are stated in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

technological adoption, thereby facilitating the production of further energy-efficient devices. Interestingly, emerging economies such as China, Mexico, India, Turkey etc. are gradually becoming major hubs of production of communication devices and further increasing the pace of ICT adoption. With the widespread adoption of ICT, emerging economies are witnessing major overhauls in lifestyle, such as working from home, online shopping, using digital media rather than printed books or newspapers, etc. All these micro-level changes due to ICT are ultimately resulting in less automotive emission and reduced anthropogenic ecological footprints at the global level.

Amongst other variables, the coefficient results reveal that GDP per capita increases on environmental degradation in the panel of 10 emerging countries. It shows that initially with rapid industrialization combined with accelerated economic growth environmental deterioration is accelerated due to scale effect [82]. According to the EKC hypothesis, economic growth contributes towards higher emissions as an economy becomes more industrialized. Industrialization, in particular, necessitates a large-scale consumption of energy resources, which results in significant environmental damage. Further, the GDPsq per capita result shows a negative influence of economic growth on emission, highlighting that the adoption of an environmentally sustainable economic growth model at a higher per capita income level helps in the advancement of the technique effect. With consistent economic growth, the economy enters a post-industrialization period, during which it reduces its emissions as a result of its economic structure and stringent environmental laws and regulations [83,84]. The inverted U-shape EKC relationship results in this study are aligned with the prior studies of [85]for Pakistan [86], for 14 Asian countries [87], for Turkey [88], for South Asian [90], for China [89], for South Asian economies among others.

The results of urbanization suggest a positive influence on CO2e, which is aligned with the results of [91], for West Africa [92], for Asian economies among others. The extant literature highlights that urbanization promotes non-renewable energy consumption

Table 7
Panel Quantile regression results.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	location	scale	qtile_5	qtile_25	qtile_50	qtile_75	qtile_95
lnren	−0.2583***	0.0028	−0.2640***	−0.2609***	−0.2585***	−0.2559***	−0.2529***
	(0.0193)	(0.0111)	(0.0314)	(0.0228)	(0.0194)	(0.0208)	(0.0279)
lngdp	1.7847***	0.0011	1.7823***	1.7836***	1.7846***	1.7856***	1.7869***
	(0.1825)	(0.1048)	(0.2965)	(0.2153)	(0.1834)	(0.1969)	(0.2634)
Lngdpsq	−0.0847***	0.0014	−0.0877***	−0.0861***	−0.0848***	−0.0835***	−0.0820***
	(0.0100)	(0.0058)	(0.0163)	(0.0119)	(0.0101)	(0.0108)	(0.0145)
lnfdi	0.0123*	−0.0038	0.0202*	0.0159*	0.0125*	0.0089	0.0048
	(0.0072)	(0.0041)	(0.0117)	(0.0085)	(0.0072)	(0.0078)	(0.0104)
lnurban	0.8183***	0.0039	0.8102***	0.8146***	0.8180***	0.8217***	0.8259***
	(0.1320)	(0.0758)	(0.2145)	(0.1557)	(0.1326)	(0.1424)	(0.1906)
lninternet	−0.0167***	−0.0044	−0.0076	−0.0125**	−0.0164***	−0.0206***	−0.0254***
	(0.0049)	(0.0028)	(0.0081)	(0.0058)	(0.0049)	(0.0053)	(0.0071)
Constant	−10.2796***	−0.0884	−10.0954***	−10.1957***	−10.2730***	−10.3566***	−10.4536***
	(0.6678)	(0.3836)	(1.0852)	(0.7878)	(0.6709)	(0.7205)	(0.9639)
Observations	196	196	196	196	196	196	196

Note: Standard errors are presented in parentheses, ***p < 0.01, **p < 0.05, *p < 0.1.

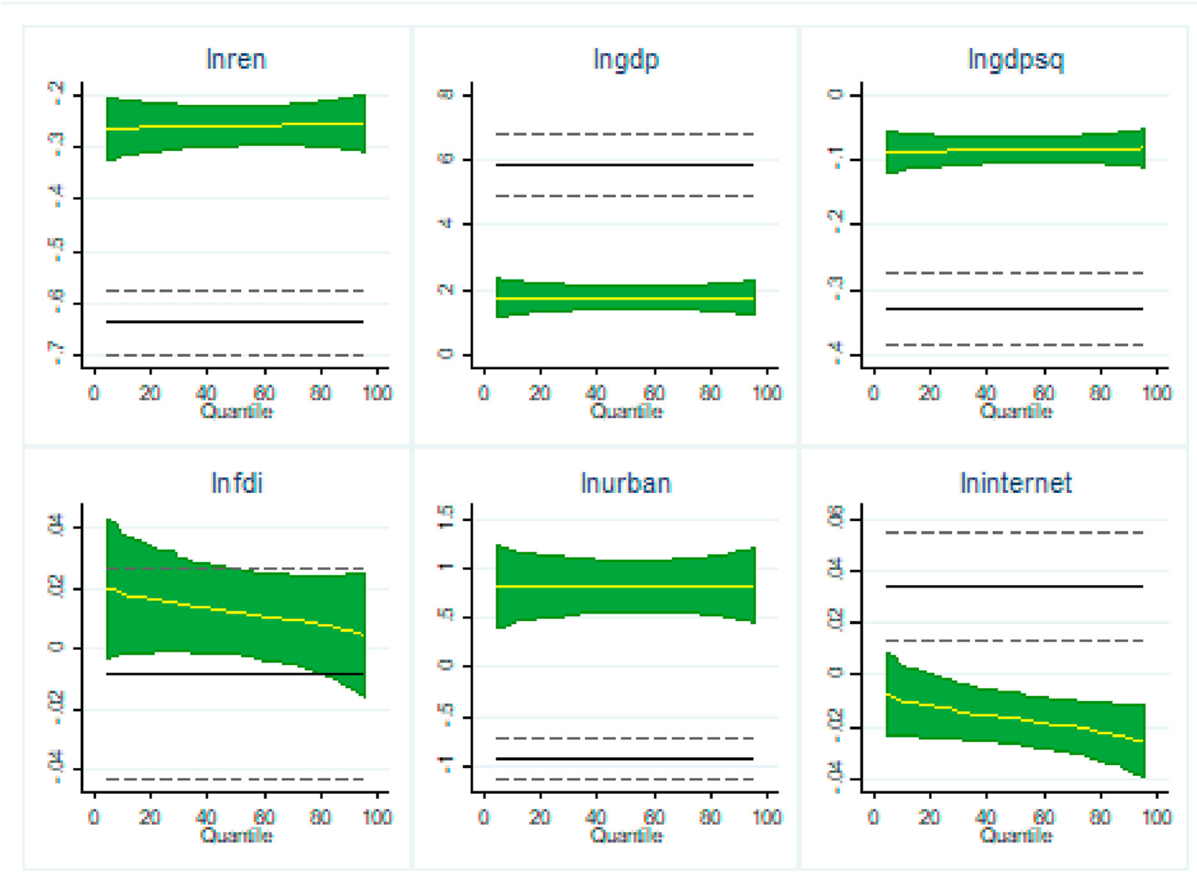

Fig. 7. MMQREG and OLS coefficient at various quantile levels, yellow lines show MMQR and solid black line show OLS based coefficients.

[93], and industrialization, with an increase in automotive emissions, thereby contributing positively towards carbon emissions.

Importantly, the results from all the regression models show a negative relationship between renewable energy and CO2e. The results are consistent with the results of [25,84] for BRICS countries [32], for China [94–96], for a panel studies, while [97] for China [98,99], for Pakistan among others. By substituting clean energy supplies for fossil fuel energy, renewable energy helps to reduce GHG emissions. As a result, the development of renewable energy is critical for emerging economies to reduce emissions. Moreover, ICT acts as a necessary enabler in this process by improving the efficiency of renewable energy devices through applications of artificial intelligence and self-optimization facilities.

5. Conclusion

The goal of the present study is to analyze the long-run impacts of information communication technology, renewable energy, and foreign direct investment within the framework of the EKC hypothesis. In doing so, the study utilized balanced panel data for emerging countries during 1996–2015. The study examined data properties through the panel unit root and panel cointegration. Due to the presence of cross-sectional dependency, a second-generation unit root test for panel data is preferred. Regression analysis for long-run relationships obtained through FMOLS, DOLS and Driscoll Kraay method supported the presence of the EKC in the sample of 10 emerging countries. For a robust analysis of the impact of exogenous variables on CO_2 emissions across various quantile levels, the study used the novel method of moments regression technique formulated by Machado and Silva [28]. Furthermore, Fig. 2 shows the variation in coefficients of MMQREG across the conditional distribution of CO2e emissions quantiles that shows supremacy over the traditional regression method. In addition, the figure shows that coefficient of renewable energy increases while moving from 0.05th-0.90th quantile. This shows that renewable energy is stronger in combating environmental damage when carbon dioxide emission is at a lower level. The coefficient of GDP is similar at all levels of quantile distribution; however, the squared term is highly effective in reducing CO2e at lower quantile levels. This shows, that once the turning point of the EKC is achieved, a very rapid reduction in environmental degradation is observed which gradually reduces. The coefficient of FDI obtained through the method of moments regression shows that it raises CO2e however, the magnitude of the coefficients tends to reduce as we move from 0.05th-0.50th quantile. Urbanization shows a consistently positive effect on CO2e at the cross all quantile levels. This shows that regardless of the level of CO2e, emerging economies are damaging the environment by urbanization. Lastly, our main focused variable internet penetration, internet use the proxy for ICT shows a negative coefficient. Interestingly, the coefficient is −0.007 which increases to become −0.02 while we move from 0.05th to 0.95th quantile levels of CO2e. Internet brings awareness about the environment; it brings efficiency in all production process along with reducing energy demand. Education and meetings are going online saving a lot of energy in transportation. Similarly, due to online commerce, the traveling and printing of papers reduce and are helpful in environmental improvement.

5.1. Policy implications

From a policy standpoint, the following steps might be taken to achieve low-carbon society. The present study proposes that renewable energy should be encouraged in emerging countries. Along with this, ICT usage, particularly internet penetration should be supported. To initiate with, innovations that cut energy use are eagerly anticipated. As a result, regulations aimed at boosting energy efficiency in the ICT industry, as well as in household and public buildings (such as the usage of smart appliances and smart grid ideas) might aid in reducing environmental deterioration. Furthermore, increasing renewable energy supply and its percentage of overall energy usage, as well as transitioning from short life cycle products and old equipment to contemporary equipment, might lessen ICT's indirect influence on CO2 emissions. Policymakers should encourage the emergence of green solutions by lowering the funding costs of environmentally friendly technology and projects. These rewards should take the form of green subsidies for the development and/or implementation of technologies. Any tax on internet usage should be either reduced or completely be eliminated to enhance its usage. For better environmental protection, internet usage must get government as well as private sector support. Finally, initiatives to raise public awareness about the advantages and environmental costs of rising ICT penetration in emerging countries should adhere to an integrated policy framework that takes into account the stage of economic development in terms of urbanization and internet penetration level.

One of the study's drawbacks is that the results and policy suggestions apply only to emerging nations and do not take into consideration the unique characteristics of each country. In truth, the growth patterns of ICT nations vary somewhat. As a result, it is critical to expanding this analysis to the national level to get a better understanding of the effect of these policies. Future research might expand on this work by finding other transmission mechanisms, especially contextual ones, via which ICT penetration affects environmental consequences other than CO_2.

CRediT authorship contribution statement

Ashar Awan: Conceptualization, Methodology, Formal analysis, Software, Data curation. **Kashif Raza Abbasi:** Formal analysis, Writing – original draft. **Soumen Rej:** Software, Validation, Writing – review & editing. **Arunava Bandyopadhyay:** Visualization, Investigation, Resources, Data curation. **Kangjuan Lv:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table A1
Cointegration Test (Kao Residual)

ADF	t-Statistic	Prob.
	−4.264170	0.0000
Residual variance	0.001287	
HAC variance	0.001072	

Table A2
Correlation Matrix

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) lnco2	1.000						
(2) lnren	−0.869	1.000					
(3) lngdp	0.729	−0.687	1.000				
(4) lngdpsq	0.713	−0.679	0.999	1.000			
(5) lnfdi	0.307	−0.264	0.214	0.207	1.000		
(6) lnurban	0.664	−0.734	0.870	0.861	0.159	1.000	
(7) lninternet	0.414	−0.270	0.584	0.587	0.279	0.405	1.000

List of abbreviations

EKC	Environmental Kuznets Curve
MMQREG	Methods of Moments Quantile Regression
ICT	Information Communication Technologies
NARDL	Nonlinear Autoregressive Distributive Lag
FDI	Foreign Direct Investment
WDI	World Development Indicators
FMOLS	Fully Modified Ordinary Least Square
DOLS	Dynamic Ordinary Least Square
CO2e	Carbon dioxide emissions
GDPsq	Gross Domestic Product square
GHG	Green House Gas

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